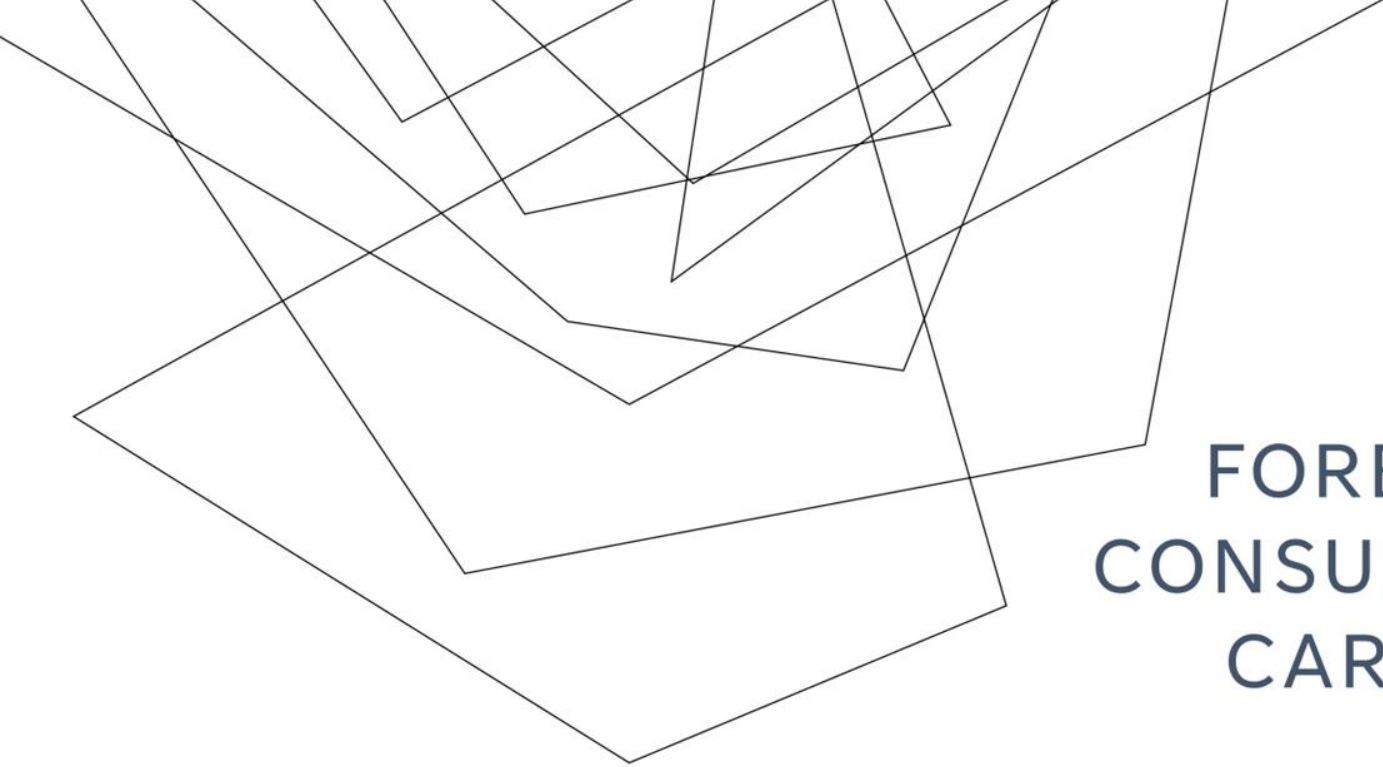


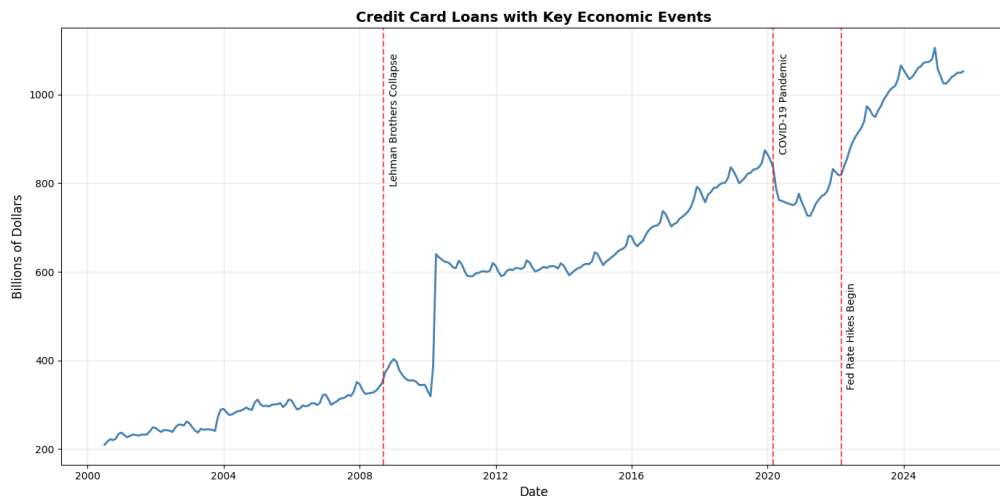
Abstract geometric lines in the top-left corner of the slide, consisting of several overlapping, irregular polygons and lines in a dark blue-grey color.

FORECASTING CONSUMER CREDIT CARD LOANS

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RESEARCH QUESTION AND MOTIVATION



The Problem:

- Consumer credit card loans = \$1.1 trillion (2024)
- Highly volatile—reacts sharply to economic shocks
- Dropped \$122 billion in the start of COVID-19

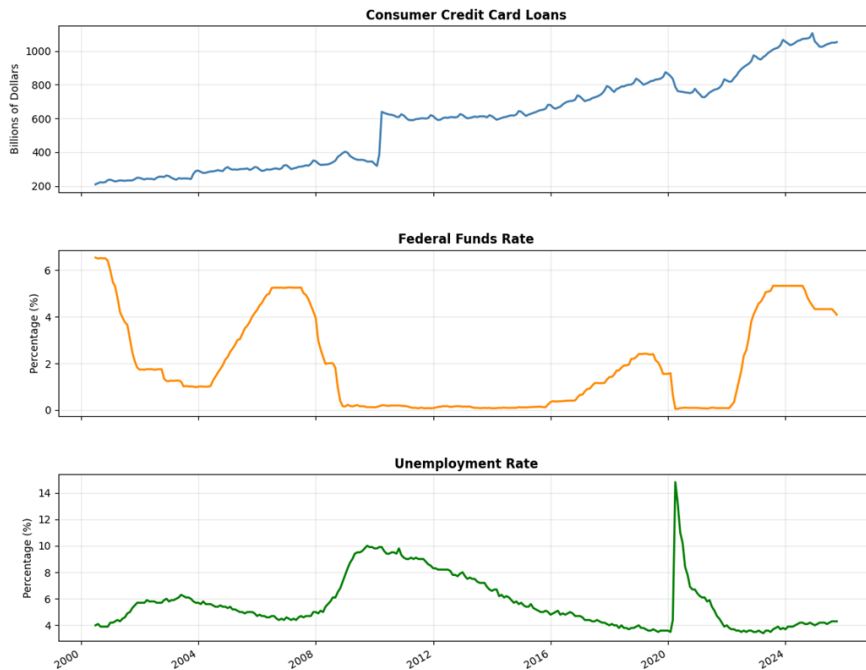
Research Question: Which time-series model best forecasts one-step-ahead monthly changes in U.S. consumer credit?

Do seasonal patterns or macroeconomic factors improve accuracy?

Why It Matters:

- ✓ Early warning signals for financial stress
- ✓ Risk management for financial institutions
- ✓ Policy insights for Federal Reserve decisions
- ✓ Understanding household financial health

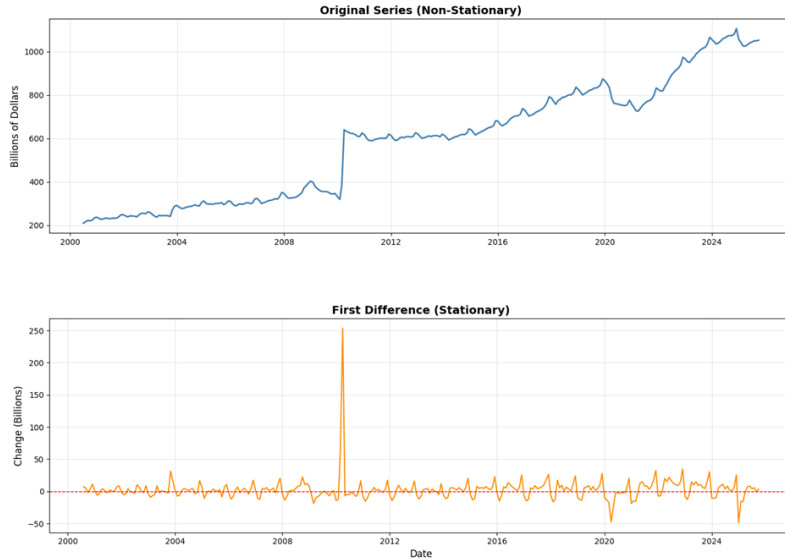
DATA OVERVIEW



Data:

- Credit card loans: 304 months (Jul 2000 - Oct 2025)
- Growth: 428% over sample period
- Exogenous variables:
 - Fed Funds Rate,
 - Unemployment Rate
- Train-Test Split: 80% training (2000-2020) | 20% test (2020-2025)
- Test period captures COVID-19 shock

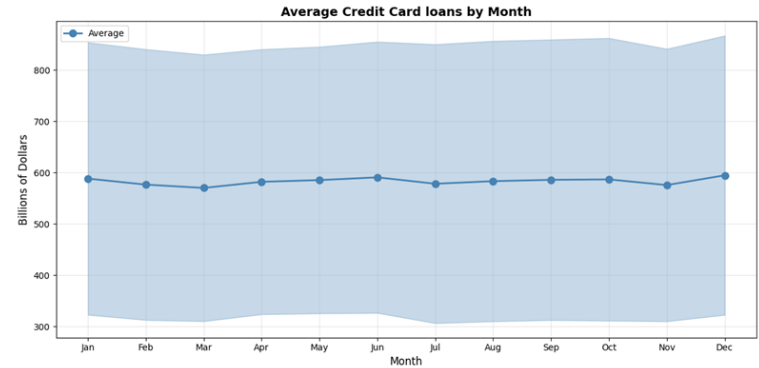
STATIONARITY & SEASONALITY



Stationarity

- Original series shows strong upward trend—not stationary
- Statistical tests confirm unit root presence ($p = 0.94$)
- First differencing removes trend, achieves stationarity

Implication: All models require $d = 1$ differencing



Seasonality

- Clear December peak driven by holiday shopping behavior
- Summer months (June) show elevated credit usage
- March dip reflects post-holiday spending pullback and tax season
- Seasonal swings demonstrate consistent annual patterns
- Implication: Seasonal ARIMA models can capture these recurring fluctuations

MODEL SPECIFICATIONS

ARIMA(0,1,1) - Baseline Model

- Captures month-to-month changes in credit card loans
- Treats all months as identical (no seasonal awareness)
- Simple one-step-ahead forecasting approach
- Serves as our benchmark for comparison
- BIC = 2110.32

SARIMA(0,1,1)(0,1,1)₁₂ - Seasonal Enhancement

- Extends ARIMA by recognizing recurring annual patterns
- Captures December holiday peaks and March tax-season lows
- Automatically adapts to predictable seasonal credit cycles
- BIC = 1904.42 (Best model fit ✓)

SARIMAX(0,1,1)(0,1,1)₁₂ - Macroeconomic Integration

- Builds on SARIMA by incorporating external economic predictors
- Includes Federal Funds Rate (monetary policy signal)
- Includes Unemployment Rate (economic health indicator)
- Tests whether macroeconomic context improves forecast accuracy
- BIC = 1908.63

ROLLING WINDOW METHODOLOGY

Real-World Forecasting Simulation:

- Generate 61 consecutive one-step-ahead forecasts (Oct 2020 - Oct 2025)
- Each model is re-estimated monthly as new data becomes available
- Mimics how forecasters operate in practice: update and predict iteratively

How the Rolling Window Works:

- Fit model using all historical data available up to month t
- Generate forecast for the next month ($t+1$)
- Wait for actual value at $t+1$ to be observed
- Expand training dataset by adding the new observation
- Re-estimate model parameters and repeat

Performance Evaluation Metrics:

- RMSE: Root Mean Squared Error – penalizes large forecast errors heavily
- MAE: Mean Absolute Error – average magnitude of errors in billions
- MAPE: Mean Absolute Percentage Error – scale-free measure for comparison

PERFORMANCE COMPARISON

Model	RMSE (\$ Billion)	MAE (\$ Billion)	MAPE (%)	BIC (Training)
ARIMA(0,1,1)	13.92	10.29	1.12	2110.32
SARIMA(0,1,1)(0,1,1) ₁₂	9.39	6.73	0.74	1904.42
SARIMAX(0,1,1)(0,1,1) ₁₂	9.40	6.75	0.74	1908.63

Key Findings:

✓ SARIMA improves 32.5% over ARIMA across all metrics

✗ SARIMAX adds no value over SARIMA (virtually identical)

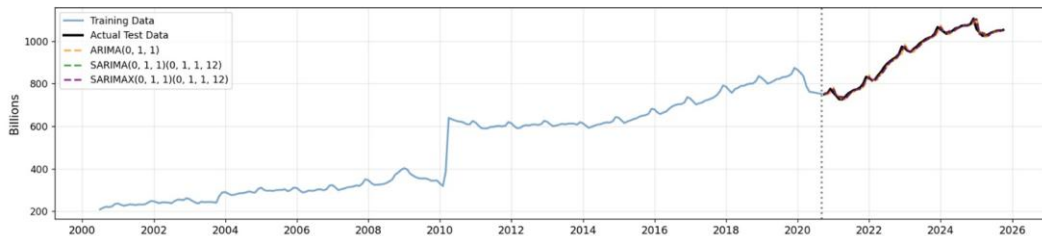
- SARIMA MAE = \$6.73B (approximately 0.7% of average credit)
- 206-point BIC improvement strongly justifies seasonal structure

Why SARIMA Is Better:

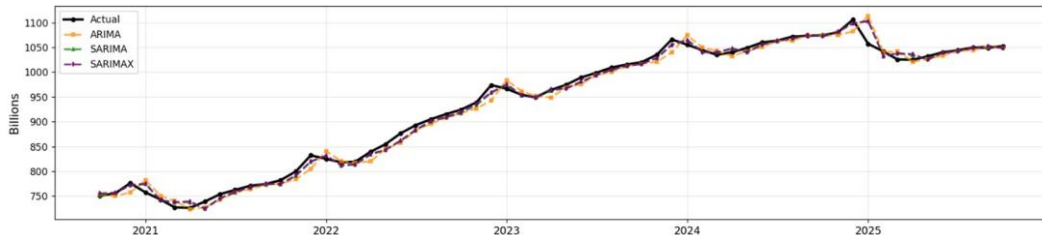
- Explicitly captures December peaks and March lows
- Adapts faster to growth rate changes
- Seasonal differencing handles year-over-year patterns

VISUAL EVIDENCE: FORECASTS VS ACTUALS

All Models: Forecast Comparison



Test Period: Detailed Model Comparison



Test Period Analysis (Oct 2020 - Oct 2025):

Phase 1: Initial Recovery (Oct 2020 - Mar 2021)

- All models underpredict as recovery accelerates faster than expected
- SARIMA/SARIMAX adapt more quickly than ARIMA

Phase 2: Rapid Surge (Apr 2021 - Dec 2022)

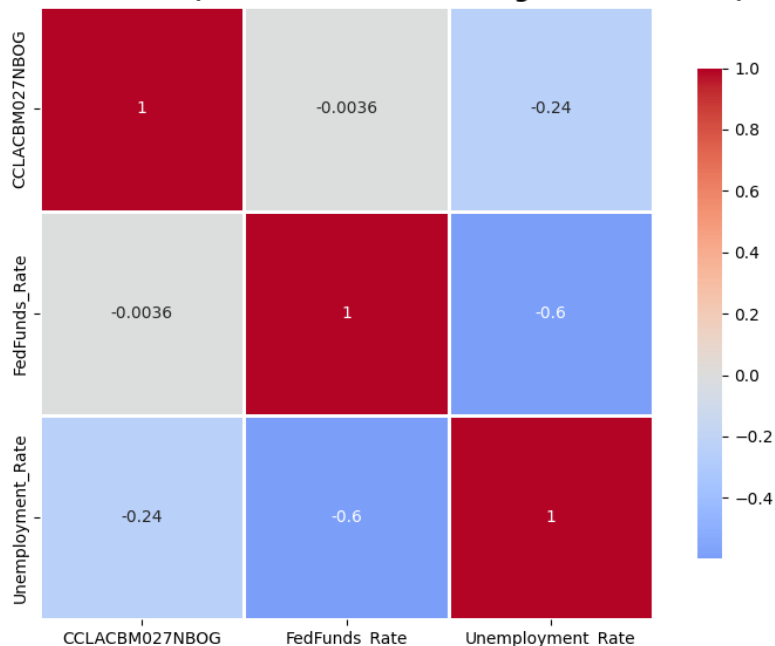
- Credit grows 34% in 21 months—unprecedented pace
- ARIMA systematically lags behind actuals throughout
- SARIMA/SARIMAX track much closer but still miss peak growth
- This period generates most forecast errors

Phase 3: Stabilization (Jan 2023 - Oct 2025)

- Growth moderates; all models improve substantially
- SARIMA/SARIMAX forecasts nearly indistinguishable from actuals

WHY SARIMAX DOESN'T IMPROVE SARIMA?

Correlation Matrix (Consumer Credits & Exogenous Variables)



Evidence:

- Fed Funds Rate: Correlation ≈ 0 , $p = 0.988$ (statistically insignificant)
- Unemployment Rate: Correlation = -0.24 , $p = 0.007$ (significant but adds no forecasting value)
- BIC penalty: 4.21 points worse than SARIMA
- Forecast accuracy: Identical to SARIMA (RMSE 9.40 vs 9.39)

Explanation:

- Information redundancy: Seasonal patterns already reflect economic conditions
- Short forecast horizon: Recent credit values dominate one-step-ahead forecasts
- Slow-moving variables: Unemployment changes gradually; effects already captured in lagged credit

Implicit information: Rapid credit growth signals strong economy without needing explicit variables

CONCLUSION

⚠ Limitations:

- **Structural breaks:** Max errors \$46.22B during 2021-2022 surge—linear models can't anticipate regime shifts
- **Point forecasts only:** No confidence intervals; uncertainty likely underestimated
- **Short horizon:** 1-month forecasts only—performance may degrade at 6-12 months
- **Sample period:** COVID-19 recovery only—untested in normal conditions
- **Linear specification:** Cannot capture threshold effects or asymmetric shocks

Main Findings:

- *Seasonality is crucial: 32.5% accuracy improvement over ARIMA*
- *Macro variables add no value: SARIMAX identical to SARIMA for 1-step forecasts*
- *SARIMA wins: Best model—accurate, parsimonious, interpretable*
- *Strong statistical justification: 206-point BIC improvement on ARIMA*